

E-AGLE

TRENTO RACING TEAM

Season 2025/2026 Report

Cockpit and Shell

Author: Pasquali Andrea

Submission Date: 24/07/2026

1. Introduction....1
1.0.1 Role in the team....1
1.0.2 Critical improvements from previous year....1
1.0.3 Report Goal and Structure....1
2. Cockpit....2
2.0.1 Project Overview and Requirements....2
2.0.2 Analog vs. Digital Architecture....2
2.0.3 First Version Design....3
2.0.4 Second Version Design....5
2.0.5 Third and Final Version Design....7
2.0.6 Adjustments applied to the final design....9
3. Steering Wheel Shell....12
3.0.1 Functionality and Schematic Adaptation....12
3.0.2 PCB Layout and Mechanical Constraints....12
Conclusions....13

1. Introduction

1.0.1 Role in the team

I joined the Eagle Electronics team in November 2024. During the early design and planning phases of the new vehicle, my initial integration involved participating in cross-functional meetings and Quality Function Deployment (QFD) sessions. I also took part in the rules quiz training sessions required for competition entry. This initial period was crucial for understanding the team's hierarchy, organization, and general workflow. Furthermore, attending internal training sessions held by Chief Technicals (CTs) and Project Managers (PMs) provided a solid technical foundation regarding the software and engineering methodologies used across different areas to develop a functional vehicle from the ground up. Following this onboarding phase, my primary role shifted to the Driver Interfaces project.

1.0.2 Critical improvements from previous year

A critical issue identified in the previous vehicle architectures was the high susceptibility of the dashboard's analog signals to electromagnetic interference (EMI) generated by the inverters. While an initial analog version of the cockpit board using hysteresis comparators and RC filters was designed, it became clear that a fundamental architectural change was necessary to ensure reliability. The primary improvement for this season was the transition to a microcontroller-based board, resolving noise issues by enabling robust CAN bus communication and expanding the system's overall versatility for additional features.

1.0.3 Report Goal and Structure

The goal of this report is to describe the design and development process of the tasks that I developed during this period. Chapter 2 focuses on the Cockpit PCB, detailing its evolution across different versions, the integration of the MCU and DC-DC converter, and the management of SCS rule-compliant LEDs and emergency signals.

Chapter 3 covers the Steering Wheel Shell, explaining its routing, component layout, and its function in providing power and managing user inputs for the main steering wheel board.

2. Cockpit

2.0.1 Project Overview and Requirements

Following the initial onboarding period, I was assigned to the Driver Interfaces project, specifically focusing on the development and design of the Cockpit PCB. The primary and fundamental role of this board is to alert the driver to various potential criticalities and display the different states of the tractive system. According to the rules, all of these indicators are classified as System Critical Signals (SCS). Throughout the development process, several additional features and on-demand functionalities were integrated to enhance the system's capabilities.

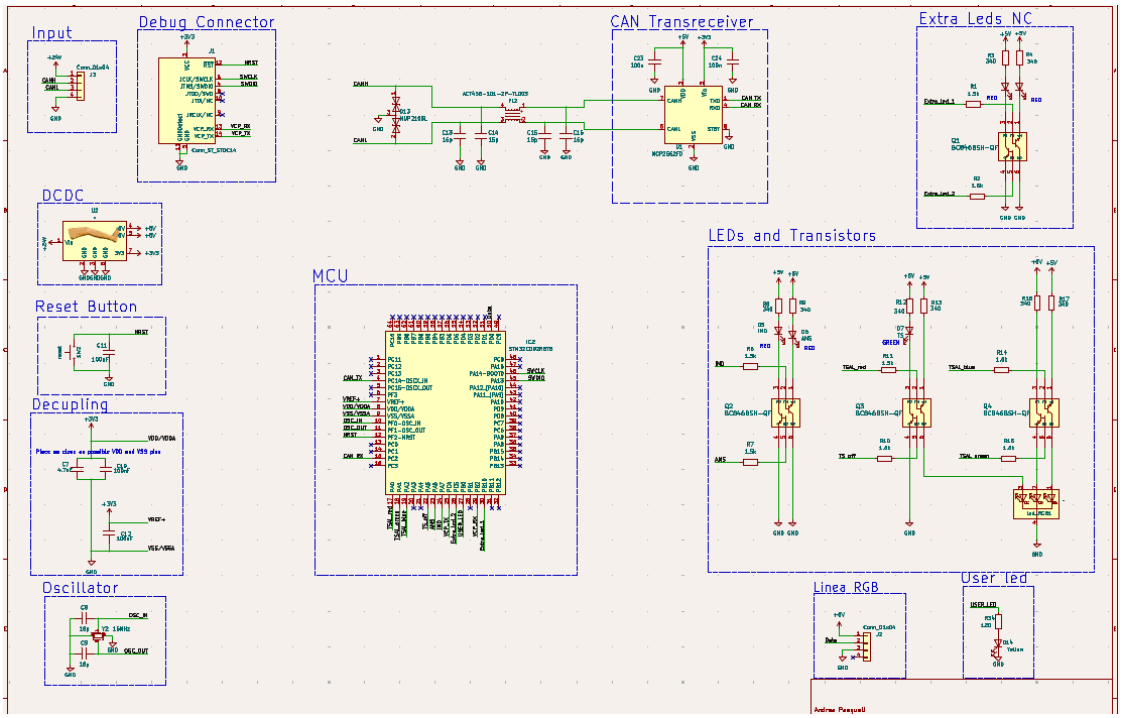
2.0.2 Analog vs. Digital Architecture

The first iteration of the Cockpit board was entirely analog. To mitigate the known issue of severe electromagnetic interference caused by the inverters, a problem also present in the team's previous vehicle, this initial design utilized hysteresis comparators and RC filters, with BJT transistors controlling the LED actuation.

However, this version had a very short lifespan. After a careful review of the regulations and consultations with other teams, the design strategy was completely revised. It was decided to introduce a microcontroller (MCU) to the board to enable CAN bus communication. This fundamental shift successfully resolved the noise-related problems while simultaneously granting the board high versatility to handle a wider range of additional functions.

2.0.3 First Version Design

The schematic of the first digital iteration was designed using KiCad, the software adopted by the team for circuit design and PCB routing.



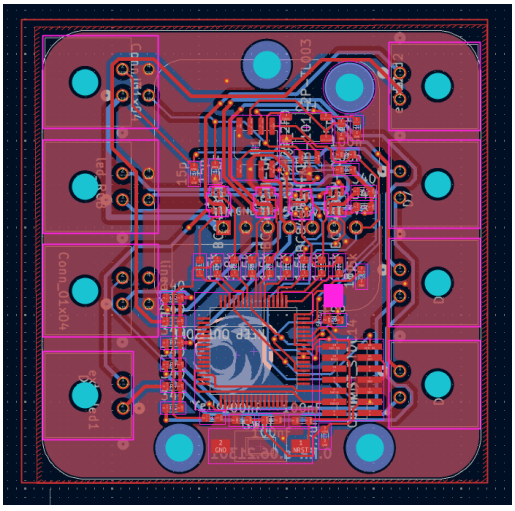
As shown in the schematic, the circuit incorporates all the essential components for the microcontroller's operation, including the debug connector, reset button, crystal oscillator for clock generation, and decoupling capacitors. Since the primary objective was to establish CAN communication, a CAN transceiver and an associated filter were included to ensure clean data exchange.

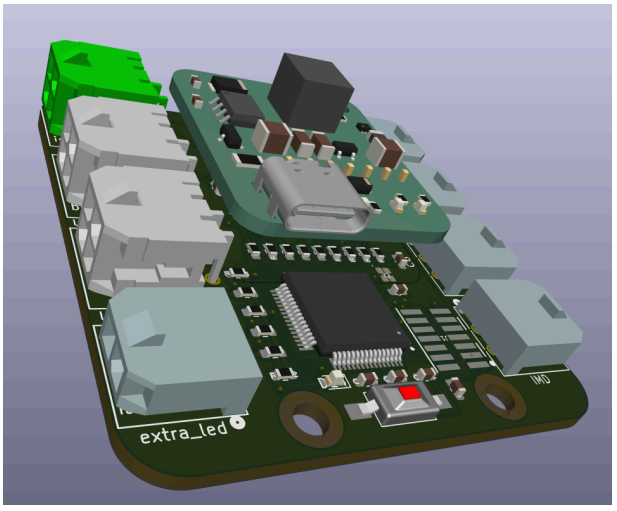
Regarding the power supply, the vehicle's Low Voltage (LV) system provides 24V from the low-voltage battery pack. However, the Cockpit PCB requires 3.3V and 5V logic levels. To step down the voltage efficiently without excessive power dissipation, a custom DC-DC converter, developed internally by other members of the electronics department was integrated into the board.

The core function of LED actuation was initially handled by BJTs, driven by the MCU pins. In this first version, the LEDs were not designed to be

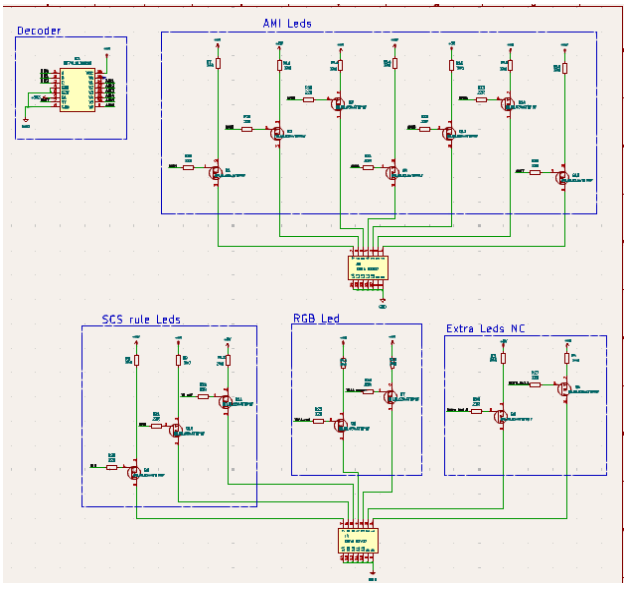
soldered directly onto the PCB. Instead, wired LEDs were used, connecting to the board via specific connectors. This modular approach allowed the LEDs to be mounted directly onto the car's chassis, providing the flexibility to house the PCB in the most convenient location within the vehicle.

Additional features included a connector for a potential addressable RGB LED strip and a dedicated RGB LED to mirror the Tractive System Active Light (TSAL) status.



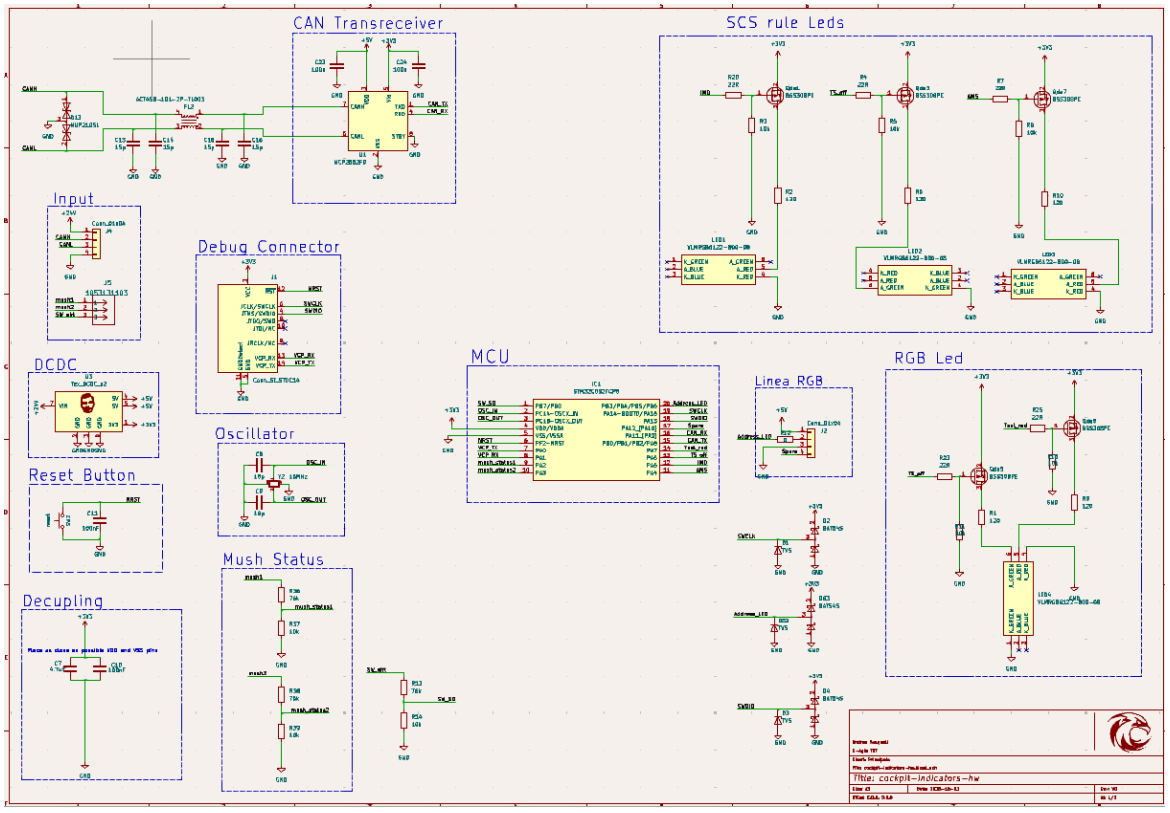


Before transitioning to the next major design revision, this first version underwent a significant modification. A decoder-based system was added to manage the Autonomous Mission Indicator (AMI) LEDs, which are also classified as SCS signals. Furthermore, the LED actuation components were upgraded from BJTs to MOSFETs, and the various output connectors were consolidated into two main macro-connectors to simplify the wiring harness.



2.0.4 Second Version Design

For the second design iteration, several key modifications were implemented to optimize both space and functionality. First, the microcontroller was replaced with a 20-pin package to minimize the number of unused pins, leading to a more efficient design.

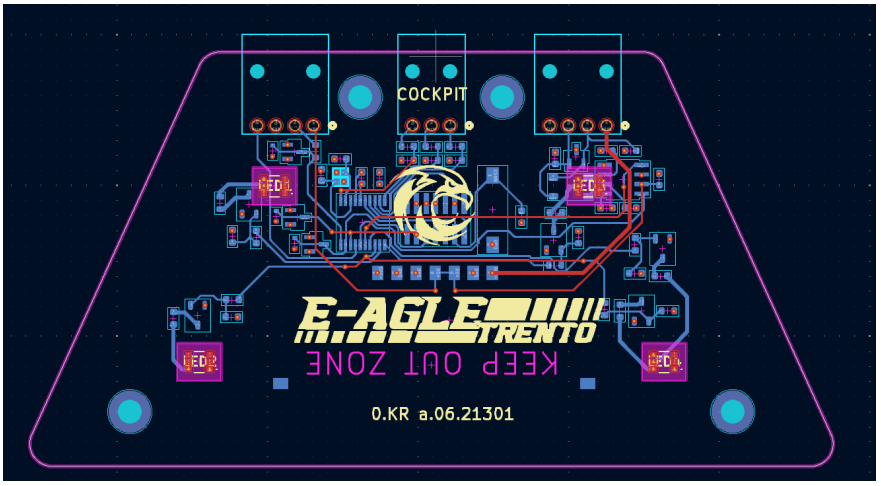


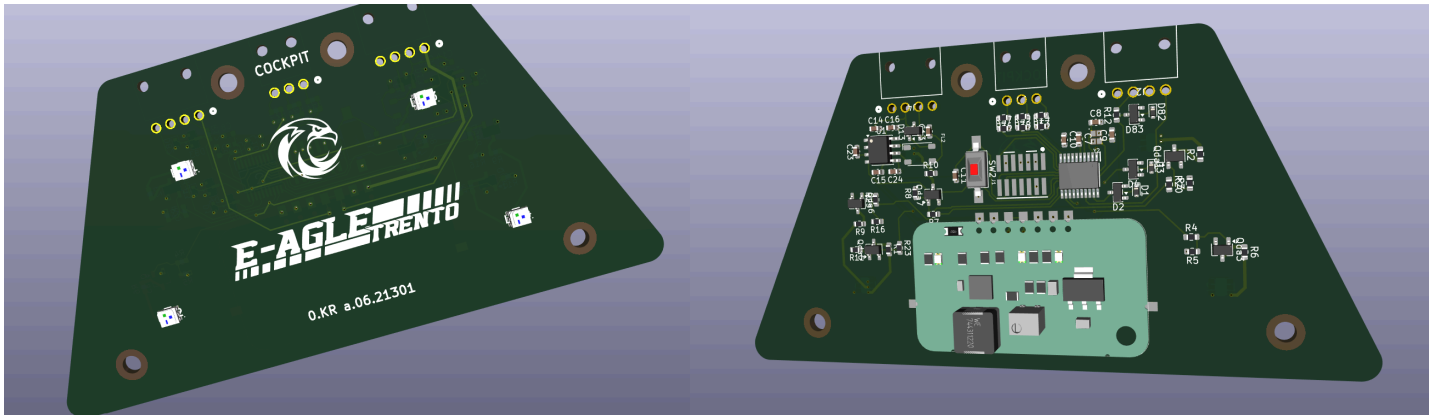
The LED footprint was also updated, transitioning to Surface Mount Device (SMD) models. The layout strategy involved placing only the LEDs on one side of the board and routing all other components on the opposite side. This double-sided component placement minimized the overall footprint, ensuring the PCB would fit seamlessly within its designated housing enclosure.

Furthermore, a significant new functionality was introduced to monitor the shutdown circuit. The board was upgraded to perform three separate voltage readings along the shutdown line. Since the shutdown line operates at a higher voltage, voltage dividers were implemented to step

down the signals to logic levels safely manageable by the MCU.

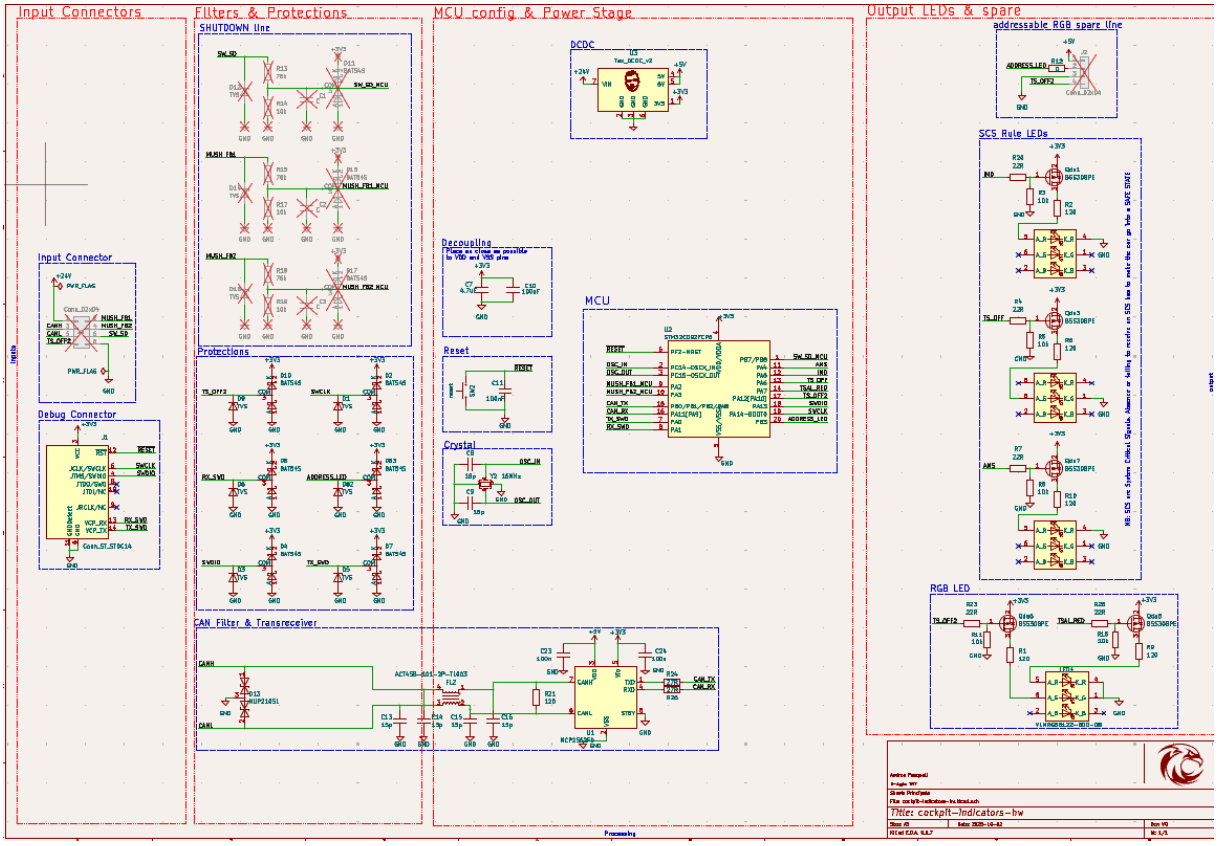
Finally, the physical outline of the PCB was completely redesigned to match the internal geometry of the dashboard container, improving the overall mechanical integration.





2.0.5 Third and Final Version Design

For the final, definitive version of the Cockpit board, the architecture was fully consolidated.



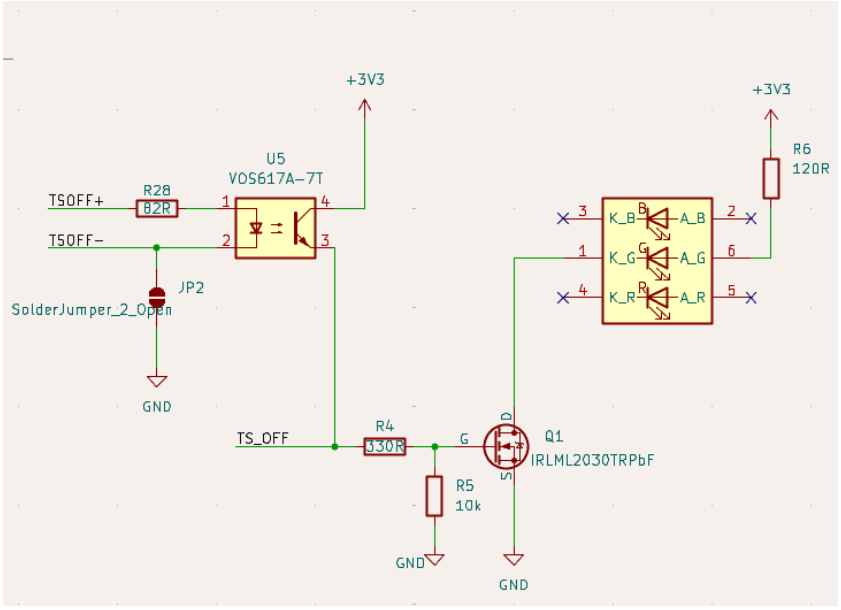
Power Management The board is powered by the vehicle's 24V supply, which is stepped down by a dedicated DC-DC converter (Tex) to provide 3.3V and 5V output rails. These voltages are necessary to power the various integrated circuits (ICs) selected for the project.

Rule Compliance and Indicators From a regulatory standpoint, the board features the mandatory LEDs: TS_off, AMS, and IMD, fully satisfying rules EV 6.3.8, EV 5.8.9, and EV 4.10.10, respectively. Additionally, an RGB LED was integrated to display the current state of the TSAL. A dedicated connector for an addressable RGB LED strip was also provided, which can serve as a backup Autonomous Mission Indicator (AMI) if needed.

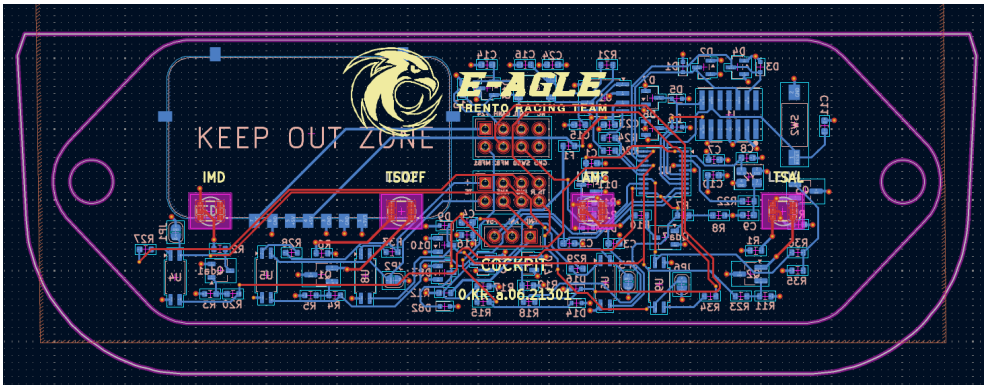
MCU Integration and Fail-Safe Actuation To definitively resolve the high signal interference issue, the microcontroller (MCU) was retained to manage LED actuation via CAN communication. The supporting circuitry includes a CAN transceiver and filter, power supply decoupling capacitors, a reset button, and a quartz crystal oscillator for clock generation. Crucially, LED actuation is controlled using the MCU's PWM outputs connected to the gates of severall MOSFETs. This specific hardware configuration guarantees a fail-safe behavior: in the event of an MCU power loss or fault state, all critical LEDs automatically turn on.

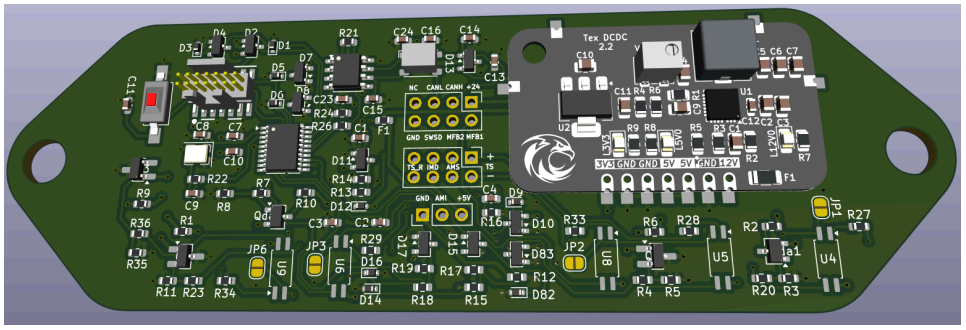
Shutdown Monitoring and Protection Furthermore, specific components were configured and placed to monitor the vehicle's shutdown circuit. This involves reading the status of two emergency stop buttons (E-stops) and the steering wheel interlock. To ensure long-term reliability, Transient Voltage Suppression (TVS) diodes were added across the microcontroller pins to protect them from external voltage spikes and transients.

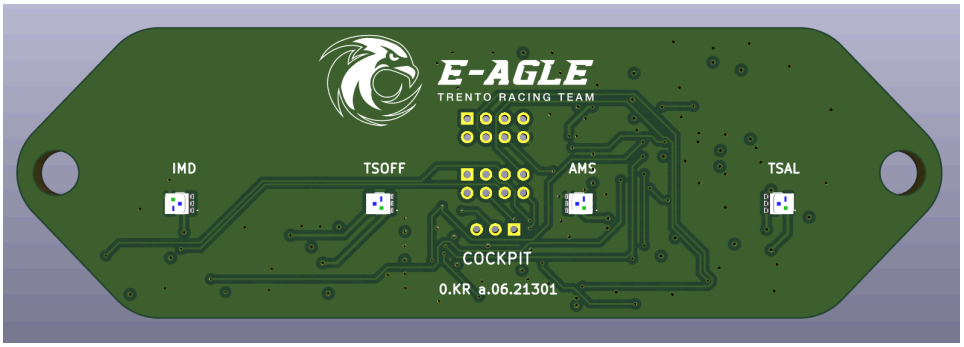
Dual-Mode Capability A major feature introduced in this final version is its dual-mode capability, allowing the board to operate in either an analog or digital configuration. This was achieved by integrating optocouplers that receive the SCS signals differentially. As a result, the board can be utilized with or without the microcontroller, offering high flexibility depending on specific operational needs or regulatory constraints.



Mechanically, the PCB outline was further resized and optimized, while preserving the same mounting zone within the dashboard for seamless integration.



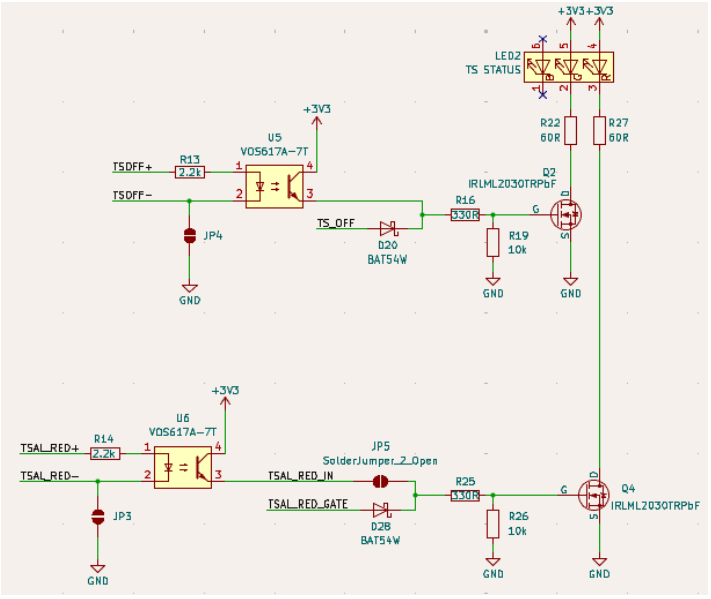




2.0.6 Adjustments applied to the final design

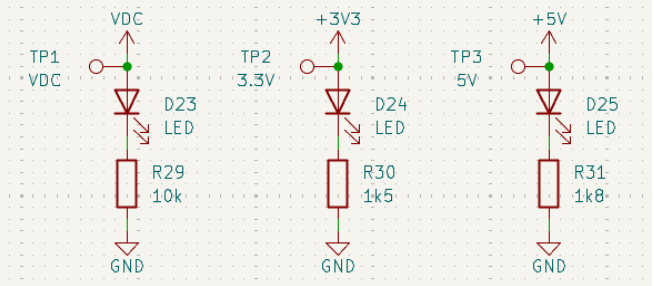
Before releasing the board for manufacturing, several minor refinements were implemented to optimize space and improve testability.

LED Consolidation and Hardware Control The dedicated TSAL status LED was removed, bringing the total number of indicators down to three. This consolidation was possible because the RGB TS STATUS LED can perform a very similar function. Additionally, as shown in the schematic below, an open solder jumper was integrated into the TSAL RED circuit line. When bridged, this jumper enables direct, analog hardware actuation of the red LED channel. If left open, the LED state is managed digitally by the MCU.

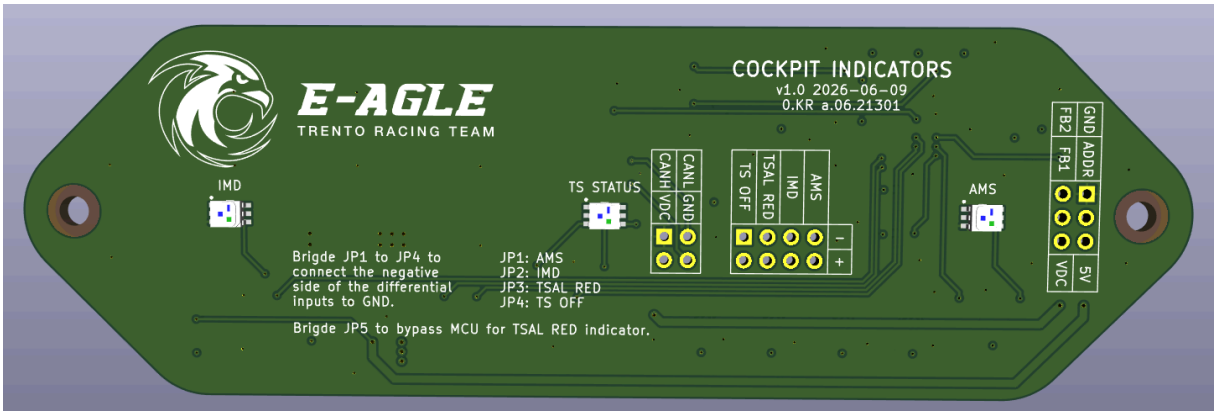


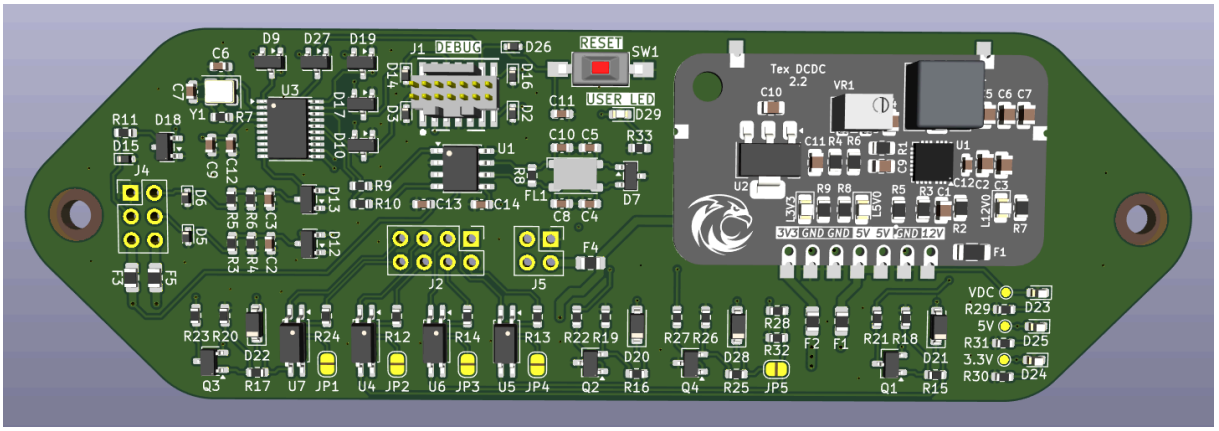
Connector Reorganization and Wiring The I/O interfaces were reorganized into three specific logical groups: one dedicated exclusively to differential signals, one for power supply and CAN communication, and a third for spare functionalities (such as the addressable RGB line and shutdown circuit voltage readings). Furthermore, due to the severe spatial constraints within the final enclosure, the physical connector headers will not be populated on the PCB. Instead, the wiring harness will be soldered directly into the through-holes of the connector footprints.

Debugging and Signal Referencing To facilitate PCB debugging and ensure correct power distribution, test points were added to all three power rails (VDC, 3.3V, and 5V). Each test point is accompanied by a dedicated status LED for immediate visual feedback.



Additionally, the board offers the flexibility to reference the analog signals directly to the common ground (GND) instead of operating in differential mode. This can be easily achieved by bridging the corresponding open solder jumpers on the input lines.





3. Steering Wheel Shell

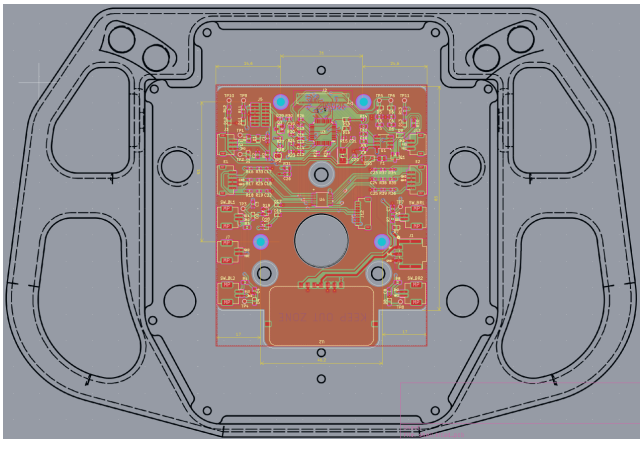
After completing the Cockpit PCB, efforts were redirected to support the finalization of the steering wheel electronics. The steering wheel architecture consists of two interconnected PCBs: a main processing board and a secondary expansion board. The work primarily focused on finalizing the design, routing, and integration of this secondary board.

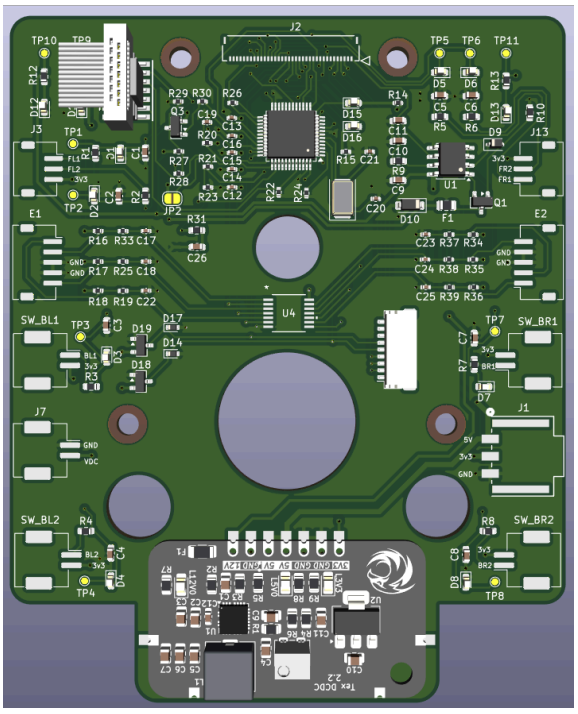
3.0.1 Functionality and Schematic Adaptation

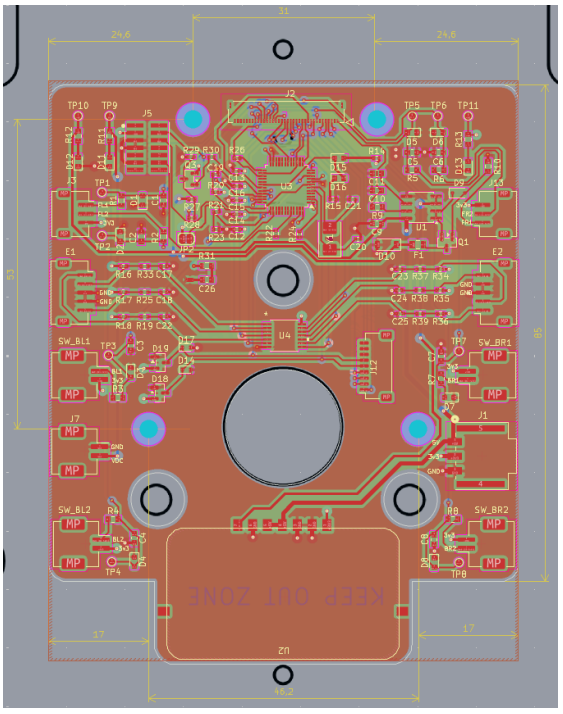
The primary function of the Shell PCB is to provide power distribution to the main board and to handle physical user inputs. It manages the signals from various push-buttons and levers (or paddles) that the driver uses to adjust steering wheel and vehicle parameters during operation. The base schematic was sourced from a public GitHub repository and was subsequently modified and tailored to meet the team's specific requirements and pinout.

3.0.2 PCB Layout and Mechanical Constraints

Regarding the PCB design and trace routing, the layout was strictly dictated by the mechanical constraints of the steering wheel assembly. The board outline and mounting holes were custom-designed to fit perfectly within the steering wheel enclosure. Furthermore, the connector placement was optimized to ensure a seamless and robust mating with its sister board. To maximize space efficiency within the tight enclosure and simplify assembly, all components were strategically placed on a single side of the PCB.







Conclusions

The development of the Driver Interfaces successfully delivered a fully integrated hardware ecosystem for the new vehicle. By transitioning the Cockpit PCB to a digital, CAN-based architecture with fail-safe logic, past electromagnetic interference vulnerabilities were effectively resolved while ensuring strict compliance with System Critical Signal (SCS) regulations. Concurrently, the finalization of the Steering Wheel Shell provided a reliable, space-optimized interface for driver inputs and power distribution. Ultimately, these upgrades resulted in a highly versatile and dependable system, ready for final vehicle integration.